① M/M/1

$\lambda = 60 h^{-1} \rightarrow \lambda = 1\ min^{-1}$

$\frac{1}{\mu} = 30s \rightarrow \mu = 2\ min^{-1}$

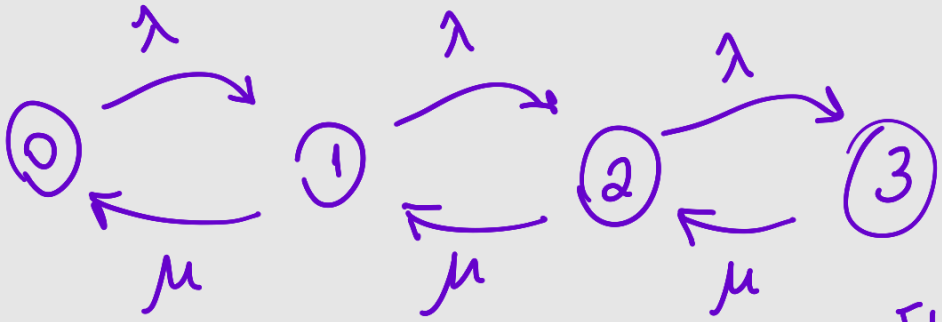
a) $\bar{n}_Q = \frac{\lambda^2}{\mu \cdot (\mu - \lambda)} = 0,5$

b) $\bar{t} = \frac{1}{\mu - \lambda}$

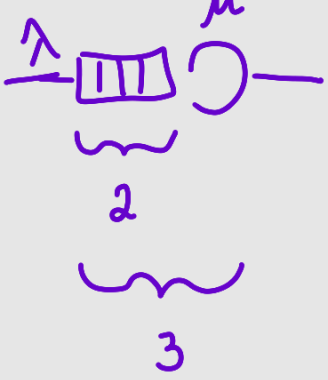
c) $P_0 = 1 - \frac{\lambda}{\mu} = 0,5 = 50\% \rightarrow 5h$

② M/M/1/3

a)



FLUXO
SAÍ = ENTRA



$P_0 = 0{,}53$

$P_1 = 0{,}265$

$P_2 = 0{,}132$

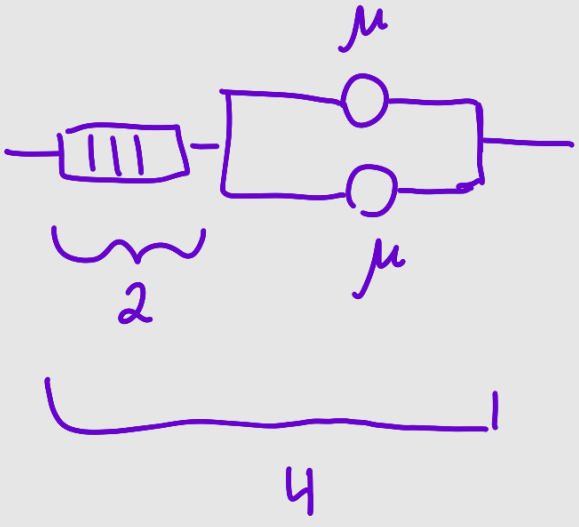
$P_3 = 0{,}066$

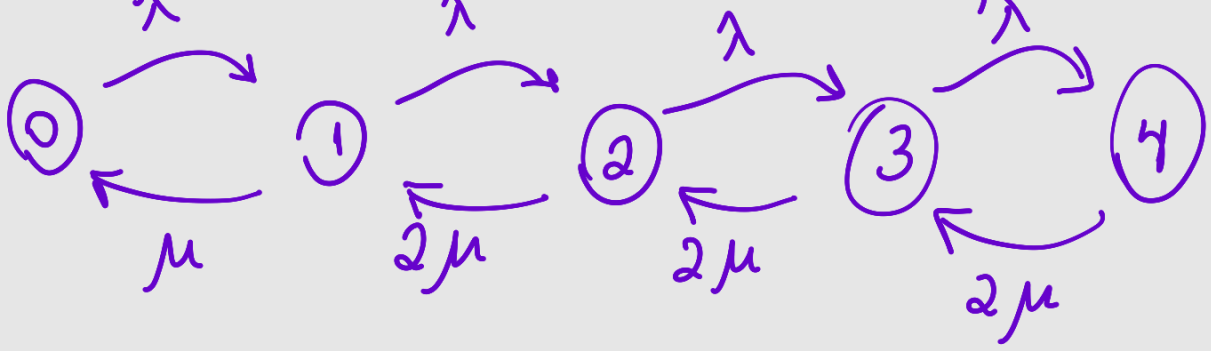
$1 - P_3 = 0{,}933 \cong 94\%$

b) $\bar{n}_Q = \lambda \cdot \bar{t}_Q \rightarrow \bar{t}_Q = \frac{\lambda}{\bar{n}_Q}$

$\bar{n}_Q = 0 \cdot P_0 + 0 \cdot P_1 + 1 \cdot P_2 + 2 \cdot P_3$

③ M/M/2/3





$P_0 = 0,6$

$P_1 = 0,3$

$P_2 = 0,075$

$P_3 = 0,0186$

$P_4 = 0,0045$

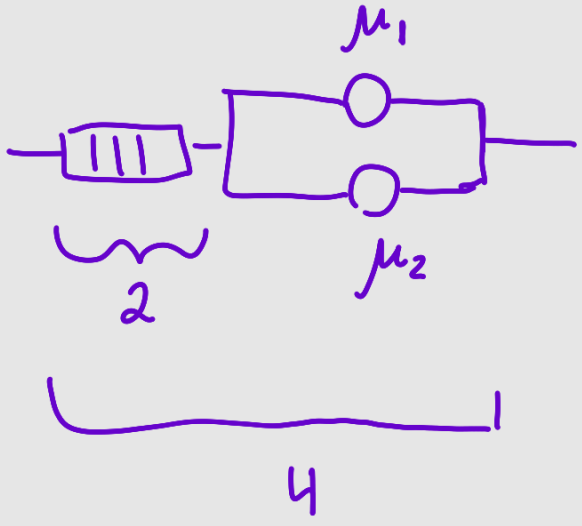
Resposta: P_4

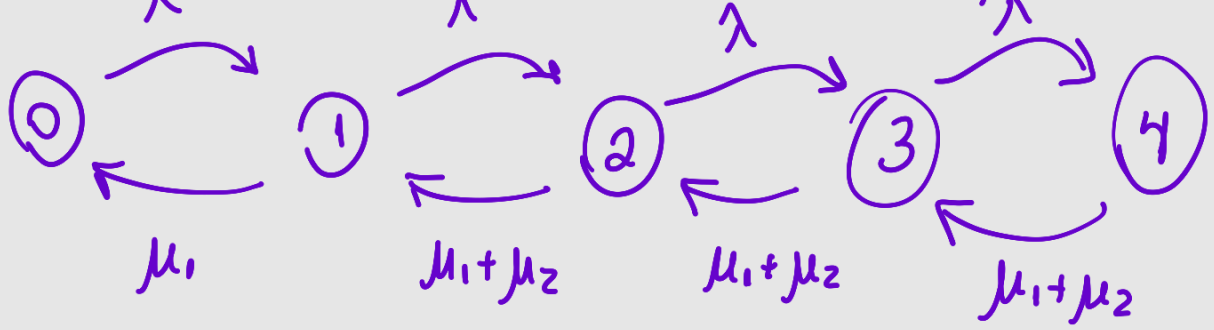
↓

0,45%

b) $\bar{T}_Q = 1,165s$

(4)





$P_2 + P_3$

$P_0 = 0,57$

$P_1 = 0,285$

$P_2 = 0,094$

$P_3 = 0,031$

$P_4 = 0,01$